

Free and Open Source Tools for Bioinformatics and Molecular Biology

The entry of open source tools in the life sciences arena has proven to be a boon.

Open source tools can be used in the predictive and diagnostic fields to provide better medical treatment. Through their use in brain mapping and DNA studies, open source tools can even be used to combat crime.



Nowadays, the applications of information and communications technology (ICT) are not limited to data transmission, cloud deployments, social media, Web servers and mobile applications. Since the last decade, IT is touching every area of the social and corporate world including health and medical sciences. Most of the medical diagnosis laboratories are now equipped with advanced computerised machines to accurately diagnose and fetch the parameters of the human body. These diagnostic machines include those used for magnetic resonance imaging (MRI), computed tomography (CT), electroencephalography (EEG), etc. These systems provide a higher degree of accuracy in the analysis of the human body, assisting doctors in diagnosing the disease and thus recommending a suitable course of treatment.

In addition to diagnostic machines, software tools and libraries are also used. These software tools and applications evaluate the biological data collected from the computerised diagnostic machines. Thus, the concept of bioinformatics has evolved, which uses software tools and applications to understand the biological and medical data. These software suites make use of high performance programming languages at the back-end to process and evaluate the biological data set, leading to effective treatment.

Bioinformatics is the interdisciplinary area that integrates biology, computer science, mathematics, engineering, chemistry and statistics for advanced predictions and analytics. The field of molecular biology is also closely associated with bioinformatics for accurate analysis of biological structures. Molecular biology deals with the deep analysis of the bimolecular movements in the cells of the body along with the details of proteins, DNA, RNA and biosynthesis.

Data sets for research in medicine and biology

With the deployment of computerised machines, researchers in diagnostic and medical sciences are taking assistance from software professionals in their field so that the programming modules can be processed by these developers. Even computer scientists are now taking the interdisciplinary field of bioinformatics for their research so that their programming knowledge can be utilised for the health sciences.

There are numerous medical data sets available for research, which are released by the diagnostic laboratories so that the overall architecture and structure of medicobiological data can be analysed by software experts. The programmers working in bioinformatics can download these medical data sets and they can perform the analysis using effective algorithms.

The software tools that can be used for the analysis and evaluation of medical data for specific types of data sets are summarised below.

OpenEEG (<http://openeeg.sourceforge.net/doc/>)

OpenEEG is free and open source software that can be used for EEG signal analysis with numerous libraries as additions, including Neuroserver, BioEra, BrainBay, Brainathlon, BrainWave Viewer and EEGMIR.

EEGNET (<https://sites.google.com/site/eegnetworks/>)

This is a free and open source tool for the analysis and visualisation of EEG brain signals. It has features to visualise the brain network.

BioSig (<http://biosig.sourceforge.net/>)

BioSig is a software library under free and open source distribution with many features of biomedical signal processing. This library has excellent features to process